# EXERCISE (O-1)

## SINGLE CORRECT TYPE QUESTIONS

**1.** In triangle $ABC$, if $\sin^3 A+\sin^3 B+\sin^3 C=3\sin A\cdot\sin B\cdot\sin C$, then the triangle is

(A) Isosceles (B) right angled

(C) scalene (D) equilateral

**MTS001**

**2.** In a triangle $ABC$, if $\angle C = 105°, \angle B = 45°$ and length of side $AC = 2$ units, then the length of the side $AB$ is equal to

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $\sqrt{2}+1$ (D) $\sqrt{3}+1$

**MTS002**

**3.** In triangle $ABC$, if $2b = a + c$ and $A - C = 90°$, then $\sin B$ equals

[**Note:** All symbols used have usual meaning in triangle $ABC$.]

(A) $\frac{\sqrt{7}}{5}$ (B) $\frac{\sqrt{5}}{8}$ (C) $\frac{\sqrt{7}}{4}$ (D) $\frac{\sqrt{5}}{3}$

**MTS005**

**4.** Angles $A, B$ and $C$ of a triangle $ABC$ are in A.P. If $\frac{b}{c}=\sqrt{\frac{3}{2}}$, then $\angle A$ is equal to

[**Note:** All symbols used have usual meaning in triangle $ABC$.]

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{5\pi}{12}$ (D) $\frac{\pi}{2}$

**MTS009**

**5.** If $K$ is a point on the side $BC$ of an equilateral triangle $ABC$ and if $\angle BAK = 15°$, then the ratio of lengths $\frac{AK}{AB}$ is

(A) $\frac{3\sqrt{2}\left(3+\sqrt{3}\right)}{2}$ (B) $\frac{\sqrt{2}\left(3+\sqrt{3}\right)}{2}$

(C) $\frac{\sqrt{2}\left(3-\sqrt{3}\right)}{2}$ (D) $\frac{3\sqrt{2}\left(3-\sqrt{3}\right)}{2}$

**MTS013**

**6.** In a triangle $\tan A:\tan B:\tan C = 1:2:3$, then $a^2:b^2:c^2$ equals

(A) 5 : 8 : 9 (B) 5 : 8 : 12

(C) 3 : 5 : 8 (D) 5 : 8 : 10

**MTS017**

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**7.** In triangle $ABC$, If $s=3+\sqrt{3}+\sqrt{2}$, $3B-C=30°, A+2B=120°$, then the length of longest side of triangle is

[**Note:** All symbols used have usual meaning in triangle $ABC$.]

(A) 2 (B) $2\sqrt{2}$

(C) $2(\sqrt{3}+1)$ (D) $\sqrt{3}-1$

**MTS020**

**8.** If in a triangle $\sin A:\sin C=\sin(A-B):\sin(B-C)$, then $a^2, b^2, c^2$

[**Note:** All symbols used have usual meaning in triangle $ABC$.]

(A) are in A.P. (B) are in G.P.

(C) are in H.P. (D) are in A.G.P

**MTS016**

**9.** In an acute triangle $ABC, \angle ABC=45°, AB=3$ and $AC=\sqrt{6}$. The angle $\angle BAC$, is

(A) 60° (B) 65° (C) 75° (D) 85°

**MTS018**

**10.** In triangle $ABC$, if $AC=8, BC=7$ and $D$ lies between $A$ and $B$ such that $AD=2, BD=4$, then the length $CD$ equals

(A) $\sqrt{46}$ (B) $\sqrt{48}$ (C) $\sqrt{51}$ (D) $\sqrt{75}$

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**MTS004**

**11.** In a triangle $ABC$, $a^3+b^3+c^3=c^2(a+b+c)$

(All symbol used have usual meaning in a triangle.)

**Statement–1:** The value of $\angle C=60°$.

**Statement –2:** $\Delta ABC$ must be equilateral.

(A) Statement–1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

**MTS006**

**12.** In triangle $ABC$, If $\frac{1}{a+c}+\frac{1}{b+c}=\frac{3}{a+b+c}$ then angle $C$ is equal to

[**Note:** All symbols used have usual meaning in triangle $ABC$.]

(A) 30° (B) 45°

(C) 60° (D) 90°

**MTS007**



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**13.** The ratio of the sides of a triangle $ABC$ is $1:\sqrt{3}:2$ . Then ratio of $\angle A:\angle B:\angle C$ is

(A) 3 : 5 : 2 (B) 1 : $\sqrt{3}$ : 2

(C) 3 : 2 : 1 (D) 1 : 2 : 3

**MTS015**

**14.** Consider a triangle $ABC$ and let $a, b$ and $c$ denote the lengths of the sides opposite to vertices $A, B$ and $C$ respectively. If $a = 1, b = 3$ and $C = 60°$, then $\sin^2 B$ is equal to

[**Note:** All symbols used have usual meaning in triangle $ABC$.]

(A) $\frac{27}{28}$ (B) $\frac{3}{28}$ (C) $\frac{81}{28}$ (D) $\frac{1}{3}$

**MTS019**

**15.** In a triangle $ABC$, if $b = (\sqrt{3}-1)a$ and $\angle C = 30°$, then the value of $(A - B)$ is equal to

(All symbols used have usual meaning in a triangle.)

(A) 30° (B) 45° (C) 60° (D) 75°

**MTS008**

**16.** In a triangle $ABC$, if $a = 13, b = 14$ and $c = 15$, then angle $A$ is equal to

(All symbols used have their usual meaning in a triangle.)

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(A) $\sin^{-1}\frac{4}{5}$ (B) $\sin^{-1}\frac{3}{5}$

(C) $\sin^{-1}\frac{3}{4}$ (D) $\sin^{-1}\frac{2}{3}$

**MTS003**

**17.** $AD$ and $BE$ are the medians of a triangle $ABC$. If $AD = 4$, $\angle DAB = \frac{\pi}{6}, \angle ABE = \frac{\pi}{3}$, then area of triangle $ABC$ equals

(A) $\frac{8}{3}$ (B) $\frac{16}{3}$ (C) $\frac{32}{3}$ (D) $\frac{32}{9}\sqrt{3}$

**MTS010**

**18.** In a triangle $ABC$, if $(a+b+c)(a+b-c)(b+c-a)(c+a-b) = \frac{8a^2b^2c^2}{a^2+b^2+c^2}$ , then the triangle is

[**Note:** All symbols used have usual meaning in triangle $ABC$.]

(A) isosceles (B) right angled

(C) equilateral (D) scalene

**MTS011**



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**19.** For right angled isosceles triangle, $\frac{r}{R}=$

[**Note:** All symbols used have usual meaning in triangle $ABC$.]

(A) $\tan\frac{\pi}{12}$ (B) $\cot\frac{\pi}{12}$ (C) $\tan\frac{\pi}{8}$ (D) $\cot\frac{\pi}{8}$

**MTS012**

**20.** Let $ABC$ be a right triangle with length of side $AB = 3$ and hypotenuse $AC = 5$.

If $D$ is a point on BC such that $\frac{BD}{DC}=\frac{AB}{AC}$, then $AD$ is equal to

(A) $\frac{4\sqrt{3}}{3}$ (B) $\frac{3\sqrt{5}}{2}$ (C) $\frac{4\sqrt{5}}{3}$ (D) $\frac{5\sqrt{3}}{4}$

**MTS014**

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# EXERCISE (O-2)

## MULTIPLE CORRECT TYPE QUESTIONS

**1.** Given an acute triangle $ABC$ such that $\sin C = \frac{4}{5}$, $\tan A = \frac{24}{7}$ and $AB = 50$. Then-

(A) centroid, orthocentre and incentre of $\Delta ABC$ are collinear

(B) $\sin B = \frac{4}{5}$

(C) $\sin B = \frac{4}{7}$

(D) area of $\Delta ABC = 1200$

**MTS036**

**2.** In $\Delta ABC$, angle $A$ is $120°$, $BC + CA = 20$ and $AB + BC = 21$, then

(A) $AB > AC$ (B) $AB < AC$

(C) $\Delta ABC$ is isosceles (D) area of $\Delta ABC = 14\sqrt{3}$

**MTS037**

**3.** In which of the following situations, it is possible to have a triangle $ABC$?
(All symbols used have usual meaning in a triangle.)

(A) $(a + c - b)(a - c + b) = 4bc$

(B) $b^2\sin 2C + c^2\sin 2B = ab$

(C) $a = 3, b = 5, c = 7$ and $C = \frac{2\pi}{3}$

(D) $\cos\left(\frac{A-C}{2}\right) = \cos\left(\frac{A+C}{2}\right)$

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**MTS038**

**4.** In $\Delta ABC$, angle $A, B$ and $C$ are in the ratio $1 : 2 : 3$, then which of the following is (are) correct?
(All symbol used have usual meaning in a triangle.)

(A) Circumradius of $\Delta ABC = c$ (B) $a : b : c = 1 : \sqrt{3} : 2$

(C) Perimeter of $\Delta ABC = 3 + \sqrt{3}$ (D) Area of $\Delta ABC = \frac{\sqrt{3}}{8}c^2$

**MTS040**

**5.** In a triangle $ABC$, let $BC = 1$, $AC = 2$ and measure of angle $C$ is $30°$. Which of the following statement(s) is (are) correct?

(A) $2\sin A = \sin B$

(B) Length of side $AB$ equals $5 - 2\sqrt{3}$

(C) Measure of angle $A$ is less than $30°$

(D) Circumradius of triangle $ABC$ is equal to length of side $AB$

**MTS041**

**6.** If the lengths of the medians $AD, BE$ and $CF$ of triangle $ABC$ are 6, 8,10 respectively, then-

(A) $AD$ & $BE$ are perpendicular

(B) $BE$ and $CF$ are perpendicular

(C) area of $\Delta ABC = 32$

(D) area of $\Delta DEF = 8$

**MTS039**



For More Join: @IITJEE_Advanced

## COMPREHENSION TYPE QUESTIONS

**Paragraph for Question No. 7 to 8**

Let incircle of $\Delta ABC$ with $\angle B = \angle C$, touches $AB, BC, CA$ at $R, P, Q$ respectively. If inradius '$r$' is integer and perimeter of $\Delta ABC$ is smallest possible integer such that $\frac{2}{AR}+\frac{5}{BP}+\frac{5}{CQ}=\frac{6}{r}$ then

**7.** The value of '$r$' is

(A) 4 (B) 3 (C) 2 (D) 1

**MTS042**

**8.** Area of $\Delta ABC$ is

(A) 15 sq. unit (B) 21 sq. unit (C) 24 sq. unit (D) 27 sq. unit

**MTS043**

## MATRIX MATCH TYPE QUESTION

**9.** If symbols have usual notations in triangle $ABC$, then match the following columns:

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (A) | $r\, r_1 r_2 r_3$ | (P) | $rs^2$ |
| (B) | $r_1 r_2 r_3$ | (Q) | $4R\cos C$ |
| (C) | $r_1 + r_2 - r_3 + r$ | (R) | $\frac{1}{2Rr}$ |
| (D) | $\frac{1}{bc}+\frac{1}{ca}+\frac{1}{ab}$ | (S) | $\Delta^2$ |

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**MTS044**

**10.** If sides of a triangle $ABC$ are $a = 6$, $b = 8$ & $c = 10$, then match the following (using usual notations)

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (A) | $r$ is equal to | (P) | 4 |
| (B) | $r_1$ is equal to | (Q) | 24 |
| (C) | $R$ is equal to | (R) | 2 |
| (D) | $\Delta$ is equal to | (S) | 6 |
| | | (T) | 5 |

**MTS067**



For More Join: @IITJEE_Advanced

# EXERCISE (S)

**1.** In $\Delta ABC$, if $a = 2b$ and $A = 3B$, then the value of $\frac{2c^2}{b^2}$ is equal to

[**Note:** All symbols used have usual meaning in triangle $ABC$. ]

**MTS024**

**2.** If the angle $A, B$ and $C$ of a triangle are in an arithmetic progression and if $a, b$ and $c$ denote the lengths of the sides opposite to $A, B$ and $C$ respectively, then the value of expression $E = \left(\frac{a}{c}\sin 2C + \frac{c}{a}\sin 2A\right)$, is $\lambda$, then the value of $\lambda^2$ is

**MTS025**

**3.** The sides $a, b, c$ (taken in that order) of triangle $ABC$ are in A.P.

If $\cos\alpha = \frac{a}{b+c}, \cos\beta = \frac{b}{c+a}, \cos\gamma = \frac{c}{a+b}$ then $\tan^2\left(\frac{\alpha}{2}\right) + \tan^2\left(\frac{\gamma}{2}\right)$ is equal to $\frac{k}{6}$, then the value of $k$ is

[**Note:** All symbols used have usual meaning in triangle $ABC$. ]

**MTS023**

**4.** In $\Delta ABC$ with usual notations, if $a = 3, b = 2$ and $\cos(A - B) = \frac{12}{13}$, then area of $\Delta ABC$ is

**MTS047**

**5.** If in a $\Delta ABC$, $a = 6$, $b = 3$ and $\cos(A - B) = 4/5$, then find its area.

[**Note:** All symbols used have usual meaning in triangle $ABC$. ]

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**MTS065**

**6.** A triangle has vertices $A, B$ and $C$, and the respective opposite sides have lengths $a, b$ and $c$. This triangle is inscribed in a circle of radius $R$. If $b = c = 1$ and the altitude from $A$ to side $BC$ has length $\sqrt{\frac{2}{3}}$, then $8R^2$ equals

[**Note:** All symbols used have usual meaning in triangle $ABC$. ]

**MTS021**

**7.** A circle is inscribed in a right triangle $ABC$, right angled at $C$. The circle is tangent to the segment $AB$ at $D$ and length of segments $AD$ and $DB$ are 7 and 13 respectively. Area of triangle $ABC$ is equal to

**MTS022**

**8.** In a $\Delta ABC$ with usual notations, if $3a + 3b = 4c$ then $\cot\frac{A}{2}\cot\frac{B}{2}$ is

**MTS045**

**9.** In $\Delta ABC$ (with usual notations), if $r_1 = \Delta$ & $r_1 = r + r_2 + r_3$, then $r$ is equal to

**MTS046**

**10.** If the area of a triangle is 100 sq.cm, $r_1$ = 10 cm and $r_2$ = 50 cm, then find the value of $(b - a)$.

[**Note:** All symbols used have usual meaning in triangle $ABC$.]

**MTS066**



For More Join: @IITJEE_Advanced

## EXERCISE - JEE (Main) PYQ

**1.** If $5, 5r, 5r^2$ are the lengths of the sides of a triangle, then $r$ cannot be equal to : **[JEE (Main) 2019]**

(1) $\frac{3}{2}$ (2) $\frac{3}{4}$ (3) $\frac{5}{4}$ (4) $\frac{7}{4}$

**MTS026**

**2.** With the usual notation, in $\Delta ABC$, if $\angle A+\angle B=120°$, $a=\sqrt{3}+1$ and $b=\sqrt{3}-1$, then the ratio $\angle A:\angle B$, is : **[JEE (Main) 2019]**

(1) 7 : 1 (2) 5 : 3 (3) 9 : 7 (4) 3 : 1

**MTS027**

**3.** In a triangle, the sum of lengths of two sides is $x$ and the product of the lengths of the same two sides is $y$. If $x^2 - c^2 = y$, where $c$ is the length of the third side of the triangle, then the circumradius of the triangle is : **[JEE (Main) 2019]**

(1) $\frac{y}{\sqrt{3}}$ (2) $\frac{c}{\sqrt{3}}$ (3) $\frac{c}{3}$ (4) $\frac{3}{2}y$

**MTS028**

**4.** Given $\frac{b+c}{11}=\frac{c+a}{12}=\frac{a+b}{13}$ for a $\Delta ABC$ with usual notation. If $\frac{\cos A}{\alpha}=\frac{\cos B}{\beta}=\frac{\cos C}{\gamma}$, then the ordered triad $(\alpha,\beta,\gamma)$ has a value :- **[JEE (Main) 2019]**

(1) (3, 4, 5) (2) (19, 7, 25) (3) (7, 19, 25) (4) (5, 12, 13)

**MTS029**

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**5.** If the lengths of the sides of a triangle are in A.P. and the greatest angle is double the smallest, then a ratio of lengths of the sides of this triangle is : **[JEE (Main) 2019]**

(1) 5 : 9 : 13 (2) 5 : 6 : 7 (3) 4 : 5 : 6 (4) 3 : 4 : 5

**MTS030**

**6.** The angles $A, B$ and $C$ of a triangle $ABC$ are in A.P. and $a: b = 1:\sqrt{3}$. If $c = 4\ cm$, then the area (in sq. $cm$) of this triangle is : **[JEE (Main) 2019]**

(1) $4\sqrt{3}$ (2) $\frac{2}{\sqrt{3}}$ (3) $2\sqrt{3}$ (4) $\frac{4}{\sqrt{3}}$

**MTS031**

**7.** If a $\Delta ABC$ has vertices $A(-1,7), B(-7,1)$ and $C(5,-5)$, then its orthocentre has coordinates: **[JEE (Main) 2020]**

(1) (3, –3) (2) $\left(-\frac{3}{5},\frac{3}{5}\right)$ (3) (–3, 3) (4) $\left(\frac{3}{5},-\frac{3}{5}\right)$

**MTS032**

**8.** If in a triangle $ABC$, $AB = 5$ units, $\angle B=\cos^{-1}\left(\frac{3}{5}\right)$ and radius of circumcircle of $\Delta ABC$ is 5 units, then the area (in sq. units) of $\Delta ABC$ is : **[JEE (Main) 2021]**

(1) $10+6\sqrt{2}$ (2) $8+2\sqrt{2}$ (3) $6+8\sqrt{3}$ (4) $4+2\sqrt{3}$

**MTS033**



For More Join: @IITJEE_Advanced

**9.** In $\Delta ABC$, the lengths of sides $AC$ and $AB$ are 12 $cm$ and 5 $cm$, respectively. If the area of $\Delta ABC$ is 30 $cm^2$ and $R$ and $r$ are respectively the radii of circumcircle and incircle of $\Delta ABC$, then the value of $2R + r$ (in $cm$) is equal to _____. **[JEE (Main) 2021]**

**MTS034**

**10.** The lengths of the sides of a triangle are $10 + x^2$, $10 + x^2$ and $20 - 2x^2$. If for $x = k$, the area of the triangle is maximum, then $3k^2$ is equal to : **[JEE (Main) 2022]**

(1) 5 (2) 8 (3) 10 (4) 12

**MTS035**

**11.** In a triangle $ABC$, if $\cos A + 2\cos B + \cos C = 2$ and the lengths of the sides opposite to the angles $A$ and $C$ are 3 and 7 respectively, then $\cos A - \cos C$ is equal to **[JEE (Main) 2023]**

(1) $\frac{3}{7}$ (2) $\frac{9}{7}$ (3) $\frac{10}{7}$ (4) $\frac{5}{7}$

**MTS064**

**12.** In a triangle $ABC$, $BC = 7$, $AC = 8$, $AB = \alpha \in N$ and $\cos A = \frac{2}{3}$. If $49\cos(3C) + 42 = \frac{m}{n}$, where $gcd(m, n) = 1$, then $m + n$ is equal to _____ **[JEE (Main) 2024]**

**MTS116**

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For More Join: @IITJEE_Advanced

# EXERCISE - JEE (Advanced) PYQ

**1.** Let $ABC$ and $ABC'$ be two non-congruent triangles with sides $AB = 4$, $AC = AC' = 2\sqrt{2}$ and angle $B = 30°$. The absolute value of the difference between the areas of these triangles is

**[JEE (Advanced) 2009]**

**MTS048**

**2.** If the angle $A, B$ and $C$ of a triangle are in an arithmetic progression and if $a, b$ and $c$ denote the length of the sides opposite to $A, B$ and $C$ respectively, then the value of the expression $\frac{a}{c}\sin 2C + \frac{c}{a}\sin 2A$, is -

**[JEE (Advanced) 2010]**

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$

**MTS049**

**3.** Consider a triangle $ABC$ and let $a, b$ and $c$ denote the length of the sides opposite to vertices $A, B$ and $C$ respectively. Suppose $a = 6, b = 10$ and the area of the triangle is $15\sqrt{3}$. If $\angle ACB$ is obtuse and if $r$ denotes the radius of the incircle of the triangle, then $r^2$ is equal to

**[JEE (Advanced) 2010]**

**MTS050**

**4.** Let $ABC$ be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let $a, b$ and $c$ denote the lengths of the sides opposite to $A, B$ and $C$ respectively. The value(s) of $x$ for which $a = x^2 + x + 1, b = x^2 - 1$ and $c = 2x + 1$ is/are

**[JEE (Advanced) 2010]**

(A) $-\left(2+\sqrt{3}\right)$ (B) $1+\sqrt{3}$ (C) $2+\sqrt{3}$ (D) $4\sqrt{3}$

**MTS051**

**5.** Let $PQR$ be a triangle of area $\Delta$ with $a = 2, b = \frac{7}{2}$ and $c = \frac{5}{2}$, where $a, b$ and $c$ are the lengths of the sides of the triangle opposite to the angles at $P, Q$ and $R$ respectively. Then $\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P}$ equals

**[JEE (Advanced) 2012]**

(A) $\frac{3}{4\Delta}$ (B) $\frac{45}{4\Delta}$ (C) $\left(\frac{3}{4\Delta}\right)^2$ (D) $\left(\frac{45}{4\Delta}\right)^2$

**MTS052**

**6.** In a triangle $PQR, P$ is the largest angle and $\cos P = \frac{1}{3}$. Further the incircle of the triangle touches the sides $PQ, QR$ and $RP$ at $N, L$ and $M$ respectively, such that the lengths of $PN, QL$ and $RM$ are consecutive even integers. Then possible length(s) of the side(s) of the triangle is (are)

**[JEE (Advanced) 2013]**

(A) 16 (B) 18 (C) 24 (D) 22

**MTS053**

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**7.** In a triangle the sum of two sides is $x$ and the product of the same two sides is $y$. If $x^2 - c^2 = y$, where $c$ is a third side of the triangle, then the ratio of the in-radius to the circum-radius of the triangle is - **[JEE (Advanced) 2014]**

(A) $\frac{3y}{2x(x+c)}$ (B) $\frac{3y}{2c(x+c)}$ (C) $\frac{3y}{4x(x+c)}$ (D) $\frac{3y}{4c(x+c)}$

**MTS054**

**8.** In a triangle $XYZ$, let $x, y, z$ be the lengths of sides opposite to the angles $X, Y, Z$, respectively and $2s = x + y + z$. If $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2}$ and area of incircle of the triangle $XYZ$ is $\frac{8\pi}{3}$, then- **[JEE (Advanced) 2016]**

(A) area of the triangle $XYZ$ is $6\sqrt{6}$

(B) the radius of circumcircle of the triangle $XYZ$ is $\frac{35}{6}\sqrt{6}$

(C) $\sin\frac{X}{2}\sin\frac{Y}{2}\sin\frac{Z}{2} = \frac{4}{35}$

(D) $\sin^2\left(\frac{X+Y}{2}\right) = \frac{3}{5}$

**MTS055**

**9.** In a triangle $PQR$, let $\angle PQR = 30°$ and the sides $PQ$ and $QR$ have lengths $10\sqrt{3}$ and 10, respectively. Then, which of the following statement(s) is (are) TRUE ? **[JEE (Advanced) 2018]**

(A) $\angle QPR = 45°$

(B) The area of the triangle $PQR$ is $25\sqrt{3}$ and $\angle QRP = 120°$

(C) The radius of the incircle of the triangle $PQR$ is $10\sqrt{3} - 15$

(D) The area of the circumcircle of the triangle $PQR$ is $100\pi$.

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**MTS056**

**10.** In a non-right-angled triangle $\Delta PQR$, let $p, q, r$ denote the lengths of the sides opposite to the angles at $P, Q, R$ respectively. The median from $R$ meets the side $PQ$ at $S$, the perpendicular from $P$ meets the side $QR$ at $E$, and $RS$ and $PE$ intersect at $O$. If $p = \sqrt{3}, q = 1$, and the radius of the circumcircle of the $\Delta PQR$ equals 1, then which of the following options is/are correct ? **[JEE (Advanced) 2019]**

(A) Area of $\Delta SOE = \frac{\sqrt{3}}{12}$ (B) Radius of incircle of $\Delta PQR = \frac{\sqrt{3}}{2}(2-\sqrt{3})$

(C) Length of $RS = \frac{\sqrt{7}}{2}$ (D) Length of $OE = \frac{1}{6}$

**MTS057**

**11.** Let $x, y$ and $z$ be positive real numbers. Suppose $x, y$ and $z$ are lengths of the sides of a triangle opposite to its angles $X, Y$ and $Z$, respectively. If $\tan\frac{X}{2} + \tan\frac{Z}{2} = \frac{2y}{x+y+z}$, then which of the following statements is/are TRUE? **[JEE (Advanced) 2020]**

(A) $2Y = X + Z$ (B) $Y = X + Z$

(C) $\tan\frac{X}{2} = \frac{x}{y+z}$ (D) $x^2 + z^2 - y^2 = xz$

**MTS058**



For More Join: @IITJEE_Advanced

**12.** In a triangle $ABC$, let $AB=\sqrt{23}$, $BC=3$ and $CA=4$. Then the value of $\frac{\cot A+\cot C}{\cot B}$ is____.

**[JEE (Advanced) 2021]**

**MTS059**

**13.** Consider a triangle $PQR$ having sides of lengths $p, q$ and $r$ opposite to the angles $P, Q$ and $R$, respectively. Then which of the following statements is (are) TRUE? **[JEE (Advanced) 2021]**

(A) $\cos P \geq 1-\frac{p^2}{2qr}$

(B) $\cos R \geq \left(\frac{q-r}{p+q}\right)\cos P+\left(\frac{p-r}{p+q}\right)\cos Q$

(C) $\frac{q+r}{p}<2\frac{\sqrt{\sin Q \sin R}}{\sin P}$

(D) If $p<q$ and $p<r$, then $\cos Q>\frac{p}{r}$ and $\cos R>\frac{p}{q}$

**MTS060**

**14.** Let $PQRS$ be a quadrilateral in a plane, where $QR=1$, $\angle PQR=\angle QRS=70°$, $\angle PQS=15°$ and $\angle PRS=40°$. If $\angle RPS=\theta°$, $PQ=\alpha$ and $PS=\beta$, then the interval(s) that contain(s) the value of $4\alpha\beta\sin\theta°$ is/are **[JEE (Advanced) 2022]**

(A) $\left(0,\sqrt{2}\right)$ (B) $(1, 2)$ (C) $\left(\sqrt{2},3\right)$ (D) $\left(2\sqrt{2},3\sqrt{2}\right)$

**MTS061**

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**PARAGRAPH "I"**

Consider an obtuse angled triangle $ABC$ in which the difference between the largest and the smallest angle is $\frac{\pi}{2}$ and whose sides are in arithmetic progression. Suppose that the vertices of this triangle lie on a circle of radius 1.

**(There are two questions based on PARAGRAPH "I", the question given below is one of them)**

**15.** Let a be the area of the triangle $ABC$. Then the value of $(64a)^2$ is **[JEE (Advanced) 2023]**

**MTS062**

**16.** Then the inradius of the triangle $ABC$ is **[JEE (Advanced) 2023]**

**MTS063**



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# ANSWER KEY

## EXERCISE (O-1)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | D | D | C | C | C | A | C | A | C | C |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | C | C | D | A | C | A | D | B | C | B |

## EXERCISE (O-2)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | |
|---|---|---|---|---|---|---|---|---|---|
| Ans. | A,B,D | A,D | B,C | B,D | A,C,D | A,C,D | C | D | |
| Que. | 9 | | | 10 | | | | | |
| Ans. | A→S;B→P;C→Q;D→R | | | A→R; B→P; C→T; D→Q | | | | | |

## EXERCISE (S)

**1.** 6 **2.** 3 **3.** 4 **4.** 3 **5.** 9
**6.** 3 **7.** 91 **8.** 7 **9.** 1 **10.** 8

## EXERCISE - JEE (Main) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 4 | 1 | 2 | 3 | 3 | 3 | 3 | 3 | 15 | 3 |
| Que. | 11 | 12 | | | | | | | | |
| Ans. | 3 | 39 | | | | | | | | |

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## EXERCISE - JEE (Advanced) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 4 | D | 3 | B | C | B,D | B | A,C,D | B,C,D | B,C,D |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | | | | |
| Ans. | B,C | 2 | A,B | A,B | 1007.99 To 1008.01 | 0.24 to 0.26 | | | | |



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